



Supply Chain & Inventory Analysis Dashboard

Developed By: Muhammad Taha Imran



Filters

Date

1/1/2025

12/31/2025

Category

All

Product

All

Region

All

City

All

Store

All

Supplier

All

147.6M

Total Spend

3K

Total Orders

388K

Wastage Cost

3.91%

Stockout Rate

46.98K

Avg Order Value

←

→

On-Time Delivery vs Target

55.27%

90.00%

100.00%

0.00%

Total Spend by Store

Sweet Crust Bakery - Cli...	
Sweet Crust Bakery - Gu...	
Sweet Crust Bakery - F-7	
Sweet Crust Bakery - DHA	

Demand by Category

168K (49.64%)

152K (44.9%)

9K (2.78%)

Dry Goods

Dairy

Baking Essen...

Perishable

Total Spend by City

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Daily Stockout Trend

10%

0%

Jan 2025

Mar 2025

May 2025

Jul 2025

Sep 2025

Nov 2025

Filters

Date

1/1/2025

12/31/2025

Category

All

Product

All

Region

All

City

All

Store

All

Supplier

All

3.91%

Stockout Rate

1K

Stockout Days

9.59

Days of Stock Cover

346.49K

Quantity Ordered

339.23K

Quantity Received

←

→

Stockout Rate by Store & Product

Store	Baking Powder	Butter (Unsalted)	Chocolate Chips
Sweet Crust Bakery - Satellite Town	0.00%	4.38%	0.00%
Sweet Crust Bakery - Gulshan	0.00%	2.47%	0.00%
Sweet Crust Bakery - Gulberg	0.00%	4.66%	0.27%
Sweet Crust Bakery - F-7	0.00%	2.47%	0.00%
Sweet Crust Bakery - DHA	0.00%	2.47%	0.00%
Total	0.00%	3.38%	0.05%

Lost Sales Value by Product

Product	Lost Sales Value
Fresh Strawberries	1.1M
Fresh Milk	0.9M
Butter (Unsalted)	0.7M
Cream Cheese	0.4M
Fresh Eggs	0.3M
Chocolate Chips	0.05M

Lost Demand by Store

Store	Lost Demand
Sweet Crust Bakery - Satellite Town	2.1K
Sweet Crust Bakery - Gulshan	1.6K
Sweet Crust Bakery - Gulberg	1.2K
Sweet Crust Bakery - F-7	1.2K
Sweet Crust Bakery - DHA	1.0K

Stockout Rate by Category

Category	Stockout Rate
Perishable	22%
Dairy	5%
Dry Goods	0.5%
Baking Essentials	0%

Stockout Days Over Time

Filters

Date

1/1/2025

12/31/2025

Category

All

Product

All

Region

All

City

All

Store

All

Supplier

All

898

Total Wasted

330.5K

Total Used

0.27%

Wastage %

388K

Wastage Cost

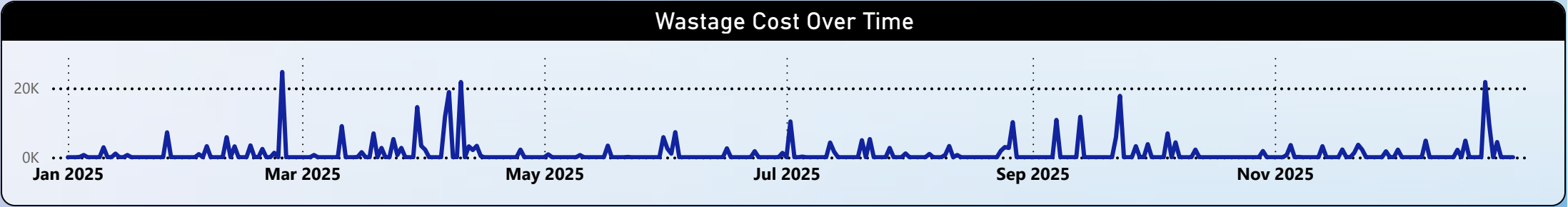
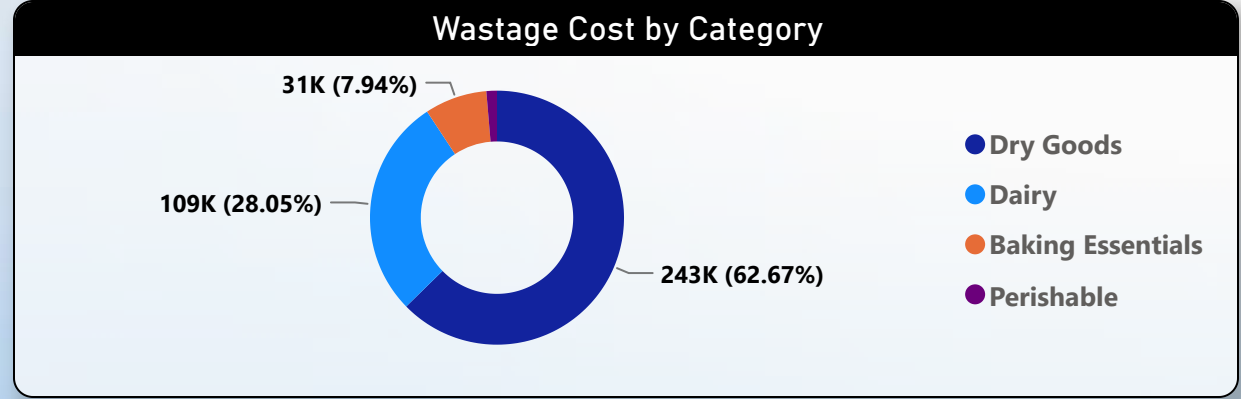
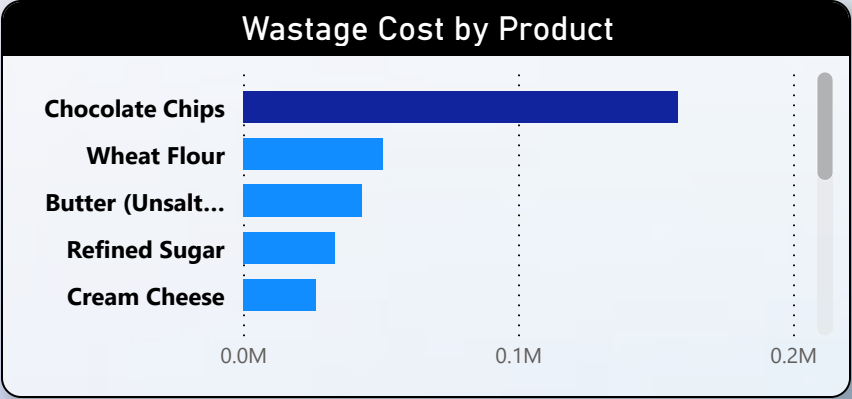
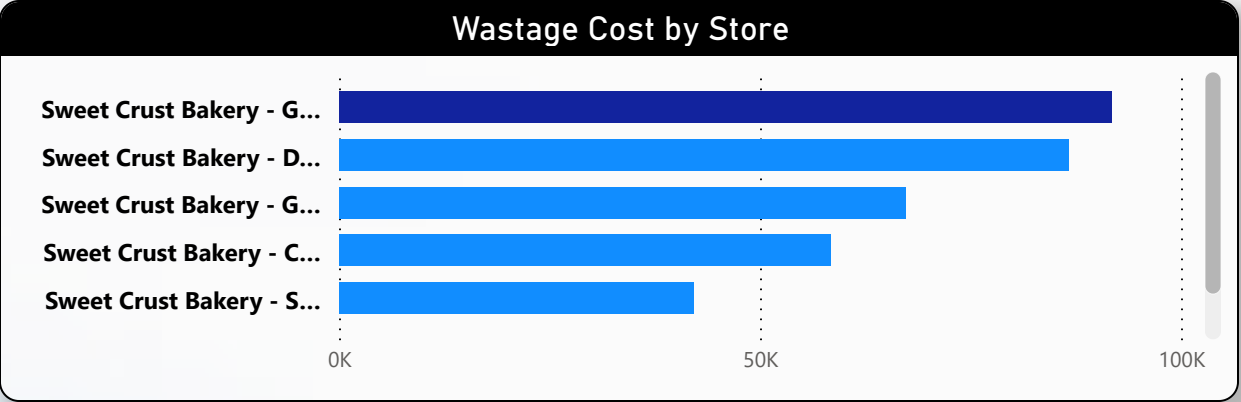
0.26%

Wastage Cost % of Spend

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Wastage Overview by Product			
Product	Shelf Life Days	Wastage %	Wastage Cost
Chocolate Chips	300	0.55%	158050
Wheat Flour	270	0.37%	50700
Butter (Unsalted)	45	0.14%	43200
Refined Sugar	365	0.33%	33300
Cream Cheese	20	0.19%	26400
Total	2858	0.27%	387500



Filters

Date

1/1/2025

12/31/2025

Category

All

Product

All

Region

All

City

All

Store

All

Supplier

All

33%
Late Delivery %

2.93
Avg Lead Time

55.27%
On-Time Delivery %

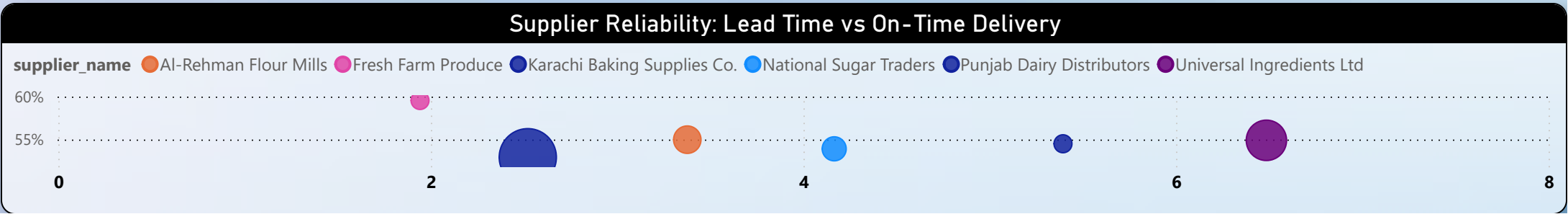
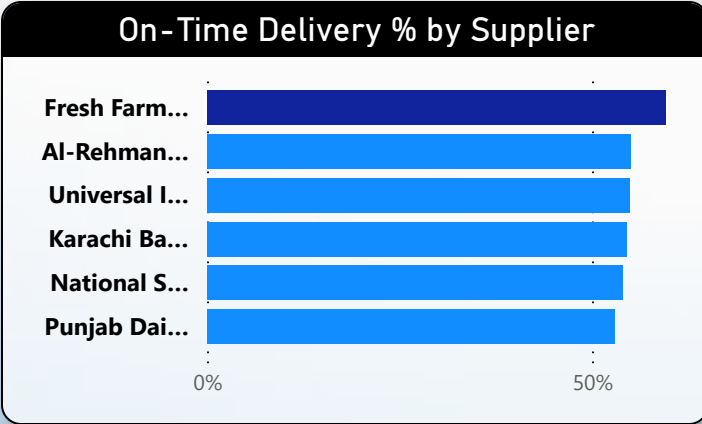
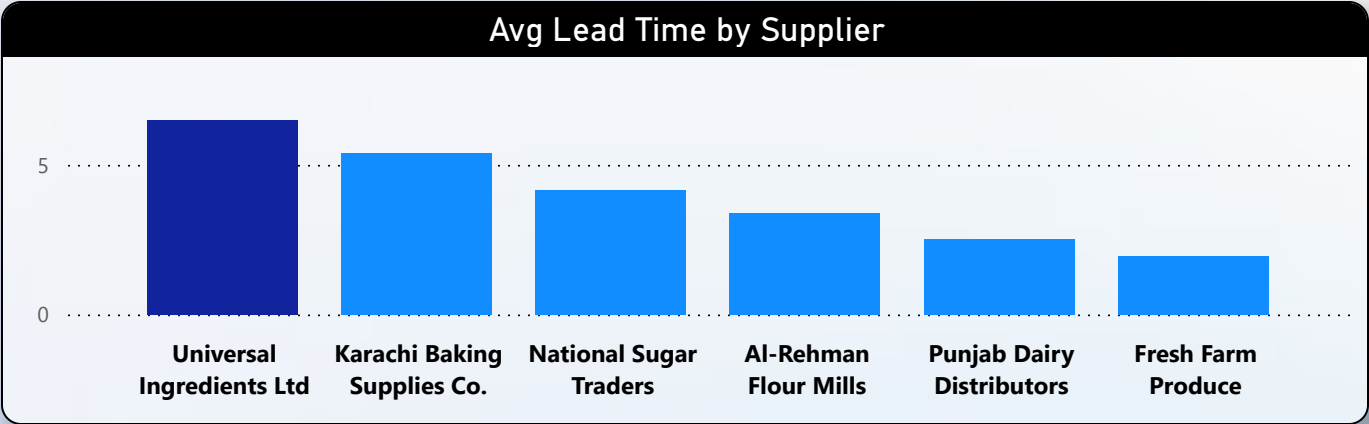
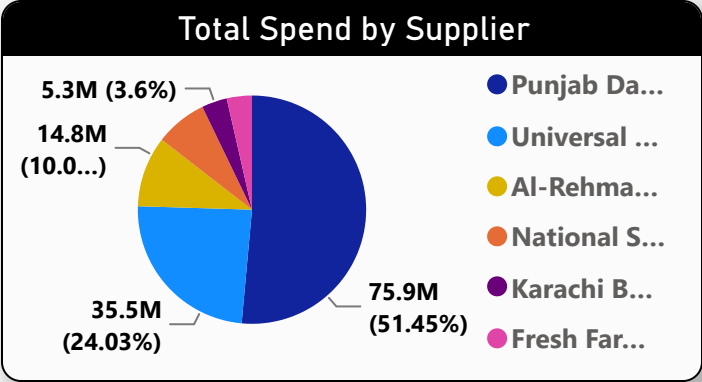
0.11
Early Delivery %

0.86
Avg Reliability Score

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Supplier Performance Summary						
Supplier Name	Reliability Score	Avg Lead Time	On-Time Delivery %	Fill Rate %	Total Spend	
Punjab Dairy Distributors	0.85	2.52	52.93%	97.37%	75917090.5	
Universal Ingredients Ltd	0.83	6.49	54.90%	97.95%	35459377.4	
Al-Rehman Flour Mills	0.92	3.38	54.98%	98.32%	14785773.3	
National Sugar Traders	0.90	4.17	53.91%	98.83%	10834749.7	
Total		2.93	55.27%	97.90%	147561320.9	



Filters

Date

1/1/2025

12/31/2025

Category

All

Product

All

Region

All

City

All

Store

All

Supplier

All

339K

Total Demand

12.91

Avg Daily Demand

4.08K

Eid Demand

29.88K

Ramadan Demand

59.59K

Wedding Season De...

←

→

Lost Demand by Product

Fresh Milk

Fresh Strawbe...

Fresh Eggs

Butter (Unsalt...

Cream Cheese

0K

5K

Total Demand by Category

9K (2.78%)

152K (44.9%)

168K (49.64%)

Dry Goods

Dairy

Baking Essentials

Perishable

Lost Demand by Store

Sweet Crust B...

Sweet Crust B...

Sweet Crust B...

Sweet Crust B...

Sweet Crust B...

0K

2K

Total Demand by Product

100K

50K

0K

Wheat Flour

Fresh Milk

Refin... Sugar

Fresh Eggs

Butter (Uns...

Choc... Chips

Cream Cheese

Fresh Stra...

Baking Pow...

Instant Yeast

Iodiz... Salt

Vanilla Extract

Monthly Demand by Product

product_name	April	August	December	February	January	July	June	March	May	November	October	September	Total
Wheat Flour	7,327.76	7,837.40	7,604.12	6,673.05	7,219.29	7,372.70	7,579.22	8,142.38	7,509.20	7,614.88	7,679.43	7,347.18	89,906.61
Fresh Milk	6,159.81	6,476.42	6,418.93	5,467.79	5,973.14	6,309.68	6,320.58	7,032.98	6,318.38	6,238.88	6,373.77	6,078.83	75,169.19
Refined Sugar	4,395.16	4,690.37	5,067.34	4,359.77	4,765.21	4,481.13	4,526.54	5,775.24	4,458.54	5,018.94	4,635.86	4,383.20	56,557.30
Fresh Eggs	3,005.54	3,104.17	3,507.94	3,116.62	3,235.27	2,987.24	3,001.17	3,645.36	2,987.67	3,487.07	3,061.69	2,923.28	38,063.02
Butter (Unsalted)	2,026.54	2,075.80	2,455.51	2,185.20	2,280.22	1,805.25	2,000.06	2,486.24	1,884.41	2,418.04	2,027.65	1,844.58	25,888.61
Total	27,137.10	28,583.16	29,980.28	26,322.48	28,091.67	27,341.10	27,639.26	32,275.92	27,438.12	29,610.45	28,075.81	26,804.47	339,299.82

Filters

Date

1/1/2025

12/31/2025

Category

All

Product

All

Region

All

City

All

Store

All

Supplier

All

147.6M

Total Spend

3K

Total Orders

46.98K

Avg Order Value

97.90%

Fill Rate %

4M

Lost Sales Value

←

→

Total Spend by Category

9.9M (6.73%)

56.5M (38.26%)

75.9M (51.45%)

Dairy

Dry Goods

Baking Essentials

Perishable

Total Spend by Supplier

10.8M (7.34%)

14.8M (10.02%)

35.5M (24.03%)

75.9M (51.45%)

Punjab Dairy Distr...

Universal Ingredie...

Al-Rehman Flour ...

National Sugar Tra...

Karachi Baking Su...

Total Spend by Product

Butter (Unsalted)

Chocolate Chips

Fresh Milk

Cream Cheese

Wheat Flour

0M

10M

20M

30M

Total Spend by Store

Sweet Crust Baker...

Sweet Crust Baker...

Sweet Crust Baker...

Sweet Crust Baker...

Sweet Crust Baker...

0M

10M

20M

30M

Total Spend Over Time

5M

0M

Jan 2025

Mar 2025

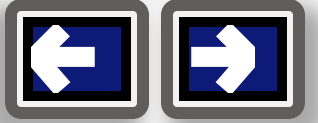
May 2025

Jul 2025

Sep 2025

Nov 2025

Key Insights



- **Only 55% of supplier deliveries arrive on time.** while 33% arrive late. This is well below a healthy target (typically 85–90%), and it's a major root cause of the stockouts seen elsewhere in the data.
- **Perishable ingredients stock out far more often than anything else.** about 28% of the time, compared to only around 5% for dairy and almost never for dry goods or baking essentials. This tells us the stockout problem is really about shelf life, not poor planning across the board.
- **Fresh Strawberries are the single biggest stockout risk.** running out roughly 1 in every 4 days. This is driven mainly by limited supply outside the winter season (Dec–Mar), not by the bakery ordering too little.
- **Wastage is concentrated in short-shelf-life items.** butter, milk, cream cheese, and strawberries account for almost all spoilage cost, while long-lasting ingredients like flour, sugar, and salt waste almost nothing.
- **Demand spikes sharply during Ramadan and Eid-ul-Fitr.** especially for sugar, butter, chocolate, and cream cheese, some ingredients see demand rise by 60–100% during this period.
- **The winter wedding season (November to February) drives a steady rise in demand.** for cake ingredients like butter, cream cheese, chocolate, and vanilla (a predictable pattern the bakery can plan stock levels around in advance).
- **Weekends are consistently busier than weekdays.** with Friday, Saturday, and Sunday demand running 15–35% higher than the rest of the week.
- **Not all suppliers are equally reliable.** Some suppliers both deliver late and short-ship the ordered quantity, meaning the bakery is dealing with two separate problems (timing and completeness) not just one.
- **Stockouts are costing real money, not just inconvenience.** Every day an ingredient runs out, that lost demand translates directly into lost sales value, since the ingredient simply couldn't be used.

Recommendations



- **Sit down with the suppliers who are late most often and find out why.** If a supplier keeps missing delivery dates, either agree on a more realistic delivery schedule with them, or start keeping a bit of extra stock on hand for the items they supply, so a late delivery doesn't turn into an empty shelf.
- **Since perishable items cause almost all the stockouts, order these more frequently and in smaller amounts** instead of trying to stock up in advance. Ingredients like flour and sugar can be ordered less often since they last much longer and rarely run out.
- **For strawberries specifically, look into a second supplier who can provide them during the off-season months** (June to October). Right now there's only one source, and that's the main reason they run out so often outside winter.
- **Review how butter, milk, and cream cheese are being ordered.** These three account for most of the spoilage cost, so ordering smaller batches more often, even if it means placing more orders, would likely save more money than it costs.
- **Increase stock levels for sugar, butter, chocolate, and cream cheese a couple of weeks before Ramadan and Eid.** Demand for these jumps sharply during this period, and running out during the busiest weeks of the year is the worst possible time to be short on stock.
- **Do the same for cake ingredients like butter, cream cheese, and vanilla heading into wedding season** (November through February). This pattern shows up every year, so it's easy to plan for ahead of time instead of reacting to it.
- **Make sure weekend stock levels are higher than weekday levels across the board,** since Friday through Sunday consistently sees more demand than the rest of the week.
- **Separate the suppliers into two groups when reviewing their performance:** those who deliver late, and those who deliver less than what was ordered. These are two different problems, and fixing one won't necessarily fix the other.
- **Treat stockouts as a cost, not just an inconvenience.** Every time an ingredient runs out, that's a direct loss, so reducing stockouts on the highest-cost items should be the first priority, since that's where the savings will be biggest.